Lecture 4
- Bayesian filter
- Graphical models
- Gaussian R.V.s
- mean, covariance

Bayesian Filter

goal: find posterior: $p(x_t | y_{1:t})$

start: $p(x_0)$

predict:

$$p(x_t | y_{1:t-1}) = \int_{x_{t-1}} p(x_t | x_{t-1}) \, p(x_{t-1} | y_{1:t-1}) \, dx_{t-1}$$

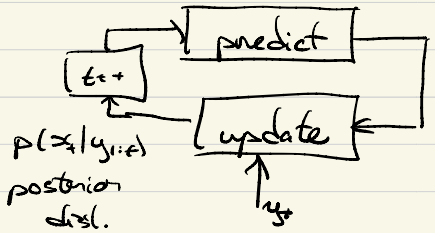
Markov trans. dist. — prior — prediction dist.

update:

$$p(x_t | y_{1:t}) = \frac{p(y_t | x_t) \, p(x_t | y_{1:t-1})}{\int_{x_t} \dots dx_t}$$

likelihood function (measurement model) — pred. dist. — posterior

algorithmic structure:



Assumptions:

(i) Cond. Indep. Measurements:

$$p(y_t | x_t, y_{1:t-1}) = p(y_t | x_t)$$

(ii) Markovianity

$$p(x_t | y_{1:t-1}, x_{0:t-1}) = p(x_t | x_{t-1})$$

proof of correctness:

predict:

$$p(x_t | y_{1:t-1}) = \int_{x_{t-1}} p(x_t, x_{t-1} | y_{1:t-1}) \, dx_{t-1} \quad \text{(marginalization)}$$

$$= \int_{x_{t-1}} p(x_t | x_{t-1}, y_{1:t-1}) \, p(x_{t-1} | y_{1:t-1}) \, dx_{t-1} \quad \text{(factorization)}$$

$$= \int_{x_{t-1}} p(x_t | x_{t-1}) \, p(x_{t-1} | y_{1:t-1}) \, dx_{t-1} \quad \text{(Markovianity)}$$

□

update:

$$p(x_t | y_{1:t}) = p(x_t | y_t, y_{1:t-1})$$

$$\text{(Bayes' Rule)} = \frac{p(y_t | x_t, y_{1:t-1}) \, p(x_t | y_{1:t-1})}{\int_{x_t} \dots dx_t}$$

$$= \frac{p(y_t | x_t) \, p(x_t | y_{1:t-1})}{\int_{x_t} \dots dx_t} \quad \text{(Cond. Indep. Meas.)}$$

□

Note: marg., factorization, Bayes' rule etc all apply to cond. dist. as well

Marg.: $p(x|z) = \int_y p(x, y | z) \, dy$

factorization: $p(x, y | z) = p(x | y, z) \, p(y | z)$

Bayes': $p(x | y, z) = \frac{p(y | x, z) \, p(x | z)}{\int_x \dots dx}$

Q: Great, we're done!

A. Not so fast!

- usually we can't compute these integrals
- special cases where we have closed form integrals
  - Histogram filter (discrete RVs)
  - Kalman filter (Gaussian RVs)
- or approximate algos or optimization based techniques
  - EKF, UKF, PF, pose graph opt.

E.g. Histogram filter (discrete R.V.)

assume X_t, Y_t discrete R.V.s

$x_t \in \{1, \dots, N\}$

$y_t \in \{1, \dots, M\}$

belief vectors:

$$\pi_t := \begin{bmatrix} p(x_t = 1 \mid y_{1:t}) \\ \vdots \\ p(x_t = N \mid y_{1:t}) \end{bmatrix}$$

so clearly $0 \le \pi_t(x_t = i) \le 1$ ← vector of 1s

and $\sum_{x_{t+1}}^{N} \pi_t(x_t) = 1 \Rightarrow 1^T \pi_t = 1$

also define pred. belief:

$$\pi_t^+ = \begin{bmatrix} p(x_t = 1 \mid y_{1:t-1}) \\ \vdots \\ p(x_t = N \mid y_{1:t-1}) \end{bmatrix}$$

Markov Transition dist. matrix

$$T = \begin{bmatrix} \vdots & & \\ \dots & p(x_t = i \mid x_{t-1} = j) & \dots \\ \vdots & & \end{bmatrix}$$

predict: $\pi_t^+ = T \pi_{t-1} \Rightarrow \pi_t^+(x_t) = \sum_{x_{t-1}} p(x_t \mid x_{t-1}) \cdot p(x_{t-1} \mid y_{1:t-1})$

update: you will find in PS 1

hint: $M(y_t) = \begin{bmatrix} p(y_t \mid x_t = 1) & & 0 \\ & \ddots & \\ 0 & & p(y_t \mid x_t = N) \end{bmatrix}$

we want: $\pi_t = f(\pi_t^+, y_t)$

E.g. Kalman Filter for Gaussian Dist. R.V.

- we will derive this over coming lectures
- we start with multi-variate Gaussian dist.

But first: Graphical Models

idea: represent conditional indeps. structure of a collection of R.V.s as a graph

Recall a graph:

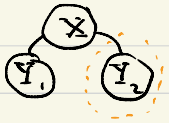
$G = (V, E)$

V: vertices "nodes"; E: edges

$V = \{X_0, X_1, X_2, \dots, X_t, Y_1, Y_2, \dots, Y_t\}$

$E = \{\dots, \{X_t, X_{t+1}\}, \{Y_t, X_t\}, \dots\}$

E.g. Cond. Indep. Measurements



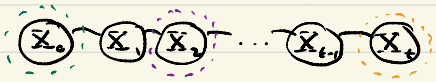
$V = \{X, Y_1, Y_2\}$

$E = \{\{X, Y_1\}, \{X, Y_2\}\}$

we say X is the "Markov blanket" for Y_2

means: $Y_2 | X$ is conditionally indep. of rest of graph

E.g. Markov Chain:



(i) $p(x_t | x_{t-1}, x_{0:t-2}) = p(x_t | x_{t-1})$

(ii) $p(x_t | x_{\neg t}) = p(x_t | x_{t-1}, x_{t+1})$

↑ all indeces except for t

(iii) $p(x_0 | x_{1:t}) = p(x_0 | x_1)$

E.g. Both Cond Indep Meas. and Markovianity

Hidden Markov Model (HMM) or Partially Observable Markov Model



notice: $p(y_t | y_{1:t-1}, x_{0:t}) = p(y_t | x_t)$

as we know

Multivariate Gaussians

recall scalar Gaussian R.V.

$X \sim \mathcal{N}(\mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\frac{(x-\mu)^2}{\sigma^2}}$

↑ mean ↑ variance

multivariate of n-dims:

$X \sim \mathcal{N}(\mu, \Sigma) = \frac{1}{\sqrt{(2\pi)^n |\Sigma|}} e^{-\frac{1}{2}(x-\mu)^T \Sigma^{-1} (x-\mu)}$

↑ det of cov. matrix

mean vector $\mu \in \mathbb{R}^n$; covariance matrix $\Sigma \in \mathbb{R}^{n \times n}$

where Σ is symmetric positive definite

so (i) $\Sigma^T = \Sigma$

(ii) $x^T \Sigma x > 0 \quad \forall x \neq 0$

notice: Σ pos. def $\Leftrightarrow \Sigma^{-1}$ pos. def.

shape of the multivariate Gaussian

